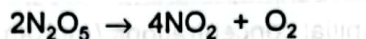


22 N_2O_5 is unstable and decomposes.



The rate equation for this reaction is shown.

$$\text{rate} = k[\text{N}_2\text{O}_5]$$

Under certain conditions, when the concentration of N_2O_5 is $0.040 \text{ mol dm}^{-3}$, the rate of the reaction is $0.00276 \text{ mol dm}^{-3} \text{ s}^{-1}$. An experiment to study this reaction under these conditions gave the following results.

time/s	$[\text{N}_2\text{O}_5] / \text{mol dm}^{-3}$
0	0.040
5	0.028
10	0.020
15	0.014
20	0.010

What are the values of k and of the half-life of this reaction?

	k / s^{-1}	half-life/s
A	0.00276	10
B	0.00276	20
C	0.0690	10
D	0.0690	20